

□ Step 1: Create Table (MySQL)

```
CREATE TABLE shape_data (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    shape_name VARCHAR(50),  
    color VARCHAR(30),  
    sides INT,  
    area DECIMAL(10,2),  
    perimeter DECIMAL(10,2),  
    category VARCHAR(20)  
);
```

□ Step 2: Insert Records

```
INSERT INTO shape_data (shape_name, color, sides, area, perimeter,  
category) VALUES  
(  
'Circle', 'Red', 0, 314.16, 62.83, 'Round'),  
(  
'Square', 'Blue', 4, 100.00, 40.00, 'Polygon'),  
(  
'Triangle', 'Green', 3, 43.30, 30.00, 'Polygon'),  
(  
'Rectangle', 'Yellow', 4, 120.00, 46.00, 'Polygon'),  
(  
'Pentagon', 'Purple', 5, 172.00, 50.00, 'Polygon'),  
(  
'Ellipse', 'Pink', 0, 200.50, 75.00, 'Round'),  
(  
'Hexagon', 'Orange', 6, 260.00, 60.00, 'Polygon'),  
(  
'Octagon', 'Gray', 8, 310.00, 80.00, 'Polygon'),  
(  
'Nonagon', 'Brown', 9, 400.00, 85.00, 'Polygon'),  
(  
'Decagon', 'Cyan', 10, 480.00, 90.00, 'Polygon'),  
(  
'Oval', 'Violet', 0, 250.00, 70.00, 'Round'),  
(  
'Rhombus', 'Indigo', 4, 150.00, 48.00, 'Polygon'),  
(  
'Trapezium', 'Magenta', 4, 175.00, 55.00, 'Polygon'),  
(  
'Parallelogram', 'Teal', 4, 200.00, 56.00, 'Polygon'),  
(  
'Kite', 'Olive', 4, 130.00, 45.00, 'Polygon'),  
(  
'Star', 'Silver', 10, 340.00, 120.00, 'Complex'),  
(  
'Heart', 'Red', 0, 280.00, 95.00, 'Curve'),  
(  
'Cube', 'Blue', 12, 600.00, 120.00, '3D'),  
(  
'Cylinder', 'Gray', 3, 750.00, 188.00, '3D'),  
(  
'Cone', 'White', 3, 400.00, 142.00, '3D'),  
(  
'Sphere', 'Black', 0, 500.00, 157.00, '3D'),  
(  
'Pyramid', 'Yellow', 5, 350.00, 105.00, '3D'),  
(  
'Tetrahedron', 'Green', 6, 300.00, 95.00, '3D'),  
(  
'Heptagon', 'Pink', 7, 220.00, 65.00, 'Polygon'),  
(  
'Dodecagon', 'Gold', 12, 520.00, 110.00, 'Polygon'),  
(  
'Semicircle', 'Orange', 0, 157.00, 40.00, 'Round'),  
(  
'Quarter Circle', 'Blue', 0, 78.50, 30.00, 'Round'),  
(  
'Annulus', 'Purple', 0, 180.00, 60.00, 'Round'),  
(  
'Crescent', 'Yellow', 0, 140.00, 50.00, 'Curve'),  
(  
'Arrow', 'Black', 5, 160.00, 55.00, 'Complex'),  
(  
'Cross', 'Gray', 12, 210.00, 85.00, 'Complex'),  
)
```

```
('Diamond', 'Cyan', 4, 145.00, 50.00, 'Polygon'),
('Lens', 'Magenta', 0, 190.00, 65.00, 'Curve'),
('Ring', 'Silver', 0, 300.00, 100.00, 'Round'),
('Prism', 'Teal', 6, 450.00, 130.00, '3D'),
('Cuboid', 'Brown', 12, 650.00, 140.00, '3D'),
('Frustum', 'White', 6, 380.00, 120.00, '3D'),
('Torus', 'Orange', 0, 700.00, 200.00, '3D'),
('Spiral', 'Violet', 0, 260.00, 90.00, 'Curve'),
('Wave', 'Blue', 0, 230.00, 85.00, 'Curve'),
('Gear', 'Gray', 20, 500.00, 150.00, 'Complex'),
('Flower', 'Pink', 0, 275.00, 95.00, 'Complex'),
('Cloud', 'White', 0, 320.00, 110.00, 'Curve'),
('Shield', 'Gold', 6, 210.00, 70.00, 'Polygon'),
('Badge', 'Red', 8, 240.00, 80.00, 'Polygon'),
('Bolt', 'Yellow', 6, 190.00, 75.00, 'Complex'),
('Drop', 'Blue', 0, 170.00, 60.00, 'Curve'),
('Leaf', 'Green', 0, 220.00, 70.00, 'Curve'),
('Egg', 'White', 0, 260.00, 75.00, 'Round'),
('Pebble', 'Gray', 0, 150.00, 55.00, 'Round');
```

Question 1: DISTINCT

□ Find all unique shape categories.

Expected Output (example):

- Polygon
 - Round
 - 3D
 - Curve
 - Complex
-

Question 2: CONCATENATE

□ Display each shape name along with its color in a single column as a description.

Expected Output (example):

- Circle is Red
- Square is Blue

- Triangle is Green
-

Question 3: ARITHMETIC OPERATORS

□ Find the total measure by adding area and perimeter for each shape.

Expected Output (example):

- CIRCLE 376.99
 - SQUARE 140.00
 - RECTANGLE 166.00
-

Question 4: COMPARISON OPERATORS

□ List all shapes having an area greater than 150.

Expected Output (example):

- PENTAGON PURPLE 5 172.00 50.00 POLYGON
 - HEXAGON ORANGE 6 260.00 60.00 POLYGON
-

Question 5: BETWEEN

□ Find all shapes whose area lies between 100 and 200.

Expected Output (example):

- SQUARE 100.00
 - RECTANGLE 120.00
 - PENTAGON 172.00
-

Question 6: IN

□ Display the shape names and colors that are either Red, Blue, or Green.

Expected Output (example):

- CIRCLE RED

- SQUARE BLUE
 - TRIANGLE GREEN
-

Question 7: LIKE

□ Find all shapes whose name ends with "gon".

Expected Output (example):

- HEXAGON
 - OCTAGON
 - NONAGON
 - DECAGON
 - PENTAGON
-

Question 8: AND / OR / NOT

□ Display all shapes that are either:

- Polygons with area > 100
- OR**
- Round shapes with perimeter < 70
- AND**
- Excluding pink ones

Expected Output (example):

- SQUARE BLUE 4 100.00 40.00 POLYGON
- CIRCLE RED 0 314.16 62.83 ROUND
- RECTANGLE YELLOW 4 120.00 46.00 POLYGON